

Iris Flower Classification using Machine Learning

Internship Project Report

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Abstract

This project classifies Iris flowers into Setosa, Versicolor, and Virginica using the K-Nearest Neighbors (KNN) algorithm. The workflow includes data loading, preprocessing, feature scaling, model training, prediction, evaluation, and deployment using Streamlit.

Objectives

- Learn supervised machine learning.
- Classify Iris flowers using KNN.
- Perform preprocessing and visualization.
- Evaluate model performance.

Dataset

The Iris dataset contains 150 samples with four numerical features: sepal length, sepal width, petal length, and petal width. There are three target classes.

Methodology

1. Load dataset
2. Exploratory Data Analysis
3. Train-Test Split (80:20)
4. StandardScaler
5. Train KNN (K=5)
6. Prediction
7. Evaluation using Accuracy, Precision, Recall, F1-Score and Confusion Matrix

Tools & Technologies

Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Streamlit, Joblib

Project Structure

train_model.py, predict.py, app.py, preprocessing.py, visualization.py, evaluation.py, dataset/, model/, outputs/, notebooks/

Results

The trained KNN model achieves high accuracy on the Iris dataset and generates evaluation metrics along with visualizations such as confusion matrix and feature plots.

Future Scope

Hyperparameter tuning, cross-validation, comparison with Decision Tree, Random Forest, Logistic Regression, and SVM, followed by cloud deployment.

Conclusion

The project demonstrates an end-to-end machine learning workflow and provides a modular, reusable implementation suitable for internship submission and future enhancement.